Week 3 assignment 1

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Grad 505

Input:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from pandas import DataFrame
from sklearn import datasets

iris = datasets.load_iris()
iris_data = DataFrame(iris.data, columns=iris.feature_names)
iris_data['species'] = iris.target_names[iris.target]
print(iris_data.head())
data = { "weight": [4.17, 5.58, 5.18, 6.11, 4.50, 4.61, 5.17, 4.53, 5.33, 5.14,
4.81, 4.17, 4.41, 3.59, 5.87, 3.83, 6.03, 4.89, 4.32, 4.69, 6.31, 5.12, 5.54,
5.50, 5.37, 5.29, 4.92, 6.15, 5.80, 5.26], "group": ["ctrl"] * 10 + ["trt1"] *
10 + ["trt2"] * 10}
PlantGrowth = pd.DataFrame(data)
print(PlantGrowth)

#1a
width =  iris_data['sepal width (cm)'].tolist()
sns.histplot(width,bins=10,kde=True)
plt.show()

#1c
mean = np.mean(width)
mean = np.round(mean,1)
median = np.median(width)
print(f"1c: mean: {mean} median: {median}")

#1d
percentile27 = np.percentile(width,[27])
print(f"1d: percentile27: {percentile27}")

#1e.
sns.scatterplot(data=iris_data,x=iris_data["sepal length (cm)"],
y=iris_data["sepal width (cm)"], hue=iris_data["species"])
plt.xlabel("sepal length (cm)")
plt.ylabel("sepal width (cm)")
plt.title("sepal length (cm) and sepal width (cm)")
plt.show()
```

```
sns.scatterplot(data=iris_data,x=iris_data["petal length (cm)"],
y=iris_data["petal width (cm)"], hue=iris_data["species"])
plt.xlabel("petal length (cm)")
plt.ylabel("petal width (cm)")
plt.title("petal length (cm) and petal width (cm)")
plt.show()

sns.scatterplot(data=iris_data,x=iris_data["sepal length (cm)"],
y=iris_data["petal length (cm)"], hue=iris_data["species"])
plt.xlabel("speal length (cm)")
plt.ylabel("petal length (cm)")
plt.title("speal length (cm) and petal length (cm)")
plt.show()

sns.scatterplot(data=iris_data,x=iris_data["sepal width (cm)"],
y=iris_data["petal width (cm)"], hue=iris_data["species"])
plt.xlabel("speal width (cm)")
plt.ylabel("petal width (cm)")
plt.title("speal width (cm) and petal width (cm)")
plt.show()

sns.scatterplot(data=iris_data, x=iris_data["species"], y=iris_data.index,
hue=iris_data["species"])
plt.xlabel("species")
plt.ylabel("index")
plt.title("species count")
plt.show()

#2a.
weight = 3.3
weight_max = 6.3
weight_range = np.arange(weight,weight_max , 0.3)
sns.histplot(data=PlantGrowth["weight"], bins=weight_range)
plt.show()

#2b
sns.boxplot(data=PlantGrowth,x="weight",hue="group")
plt.show()

#2d
trt1 = pd.DataFrame(PlantGrowth[PlantGrowth["group"] == "trt1"])
trt2 = pd.DataFrame(PlantGrowth[PlantGrowth["group"] == "trt2"])
print(trt1.count())
trt2_min = trt2['weight'].min()
print(trt2_min)
less_than = pd.DataFrame(trt1[trt1["weight"] < trt2_min])
less_than_percent = (less_than.count()/trt1.count())*100
print(f"1d: {less_than_percent}")
```

```
#2e
groups = PlantGrowth[PlantGrowth["weight"] > 5.5][["group"]]
sns.countplot(data=groups,x="group",y=None,hue="group",palette="pastel")
plt.show()
```

Output:

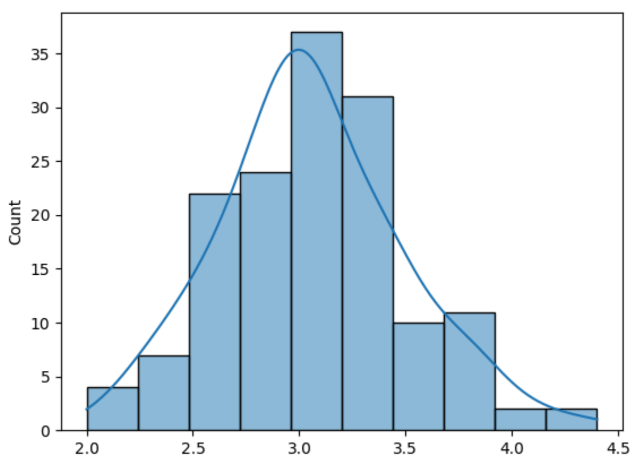
```
"/Users/ajohns/PycharmProjects/Intro to DS/.venv/bin/python"
/Users/ajohns/PycharmProjects/Intro to DS/week03assignment.py
   sepal length (cm)  sepal width (cm)  ...  petal width (cm)  species
0                5.1               3.5  ...               0.2   setosa
1                4.9               3.0  ...               0.2   setosa
2                4.7               3.2  ...               0.2   setosa
3                4.6               3.1  ...               0.2   setosa
4                5.0               3.6  ...               0.2   setosa

[5 rows x 5 columns]
    weight group
0     4.17  ctrl
1     5.58  ctrl
2     5.18  ctrl
3     6.11  ctrl
4     4.50  ctrl
5     4.61  ctrl
6     5.17  ctrl
7     4.53  ctrl
8     5.33  ctrl
9     5.14  ctrl
10    4.81  trt1
11    4.17  trt1
12    4.41  trt1
13    3.59  trt1
14    5.87  trt1
15    3.83  trt1
16    6.03  trt1
17    4.89  trt1
18    4.32  trt1
19    4.69  trt1
20    6.31  trt2
21    5.12  trt2
22    5.54  trt2
23    5.50  trt2
24    5.37  trt2
25    5.29  trt2
```

26   4.92 trt2
27   6.15 trt2
28   5.80 trt2
29   5.26 trt2
1c: mean: 3.1 median: 3.0
1d: percentile27: [2.8]
weight   10
group   10
dtype: int64
4.92
1d: weight   80.0
group   80.0
dtype: float64

Process finished with exit code 0

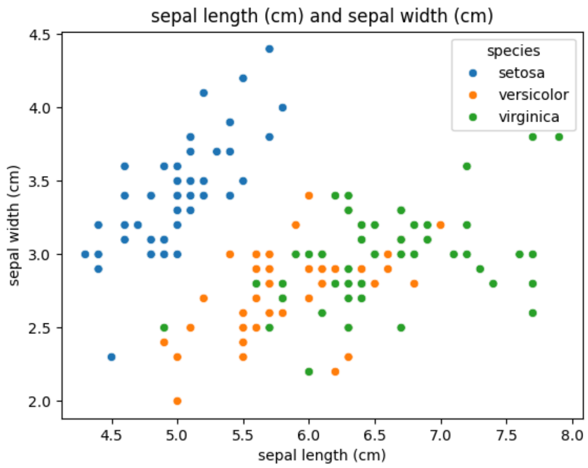
Answers:

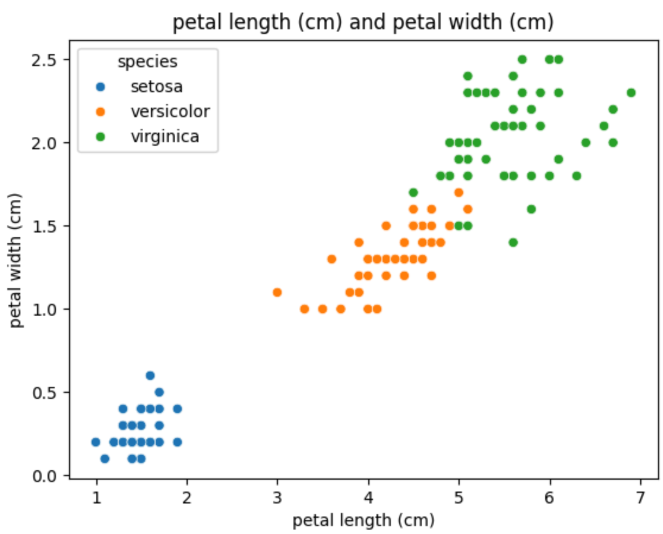
1a.

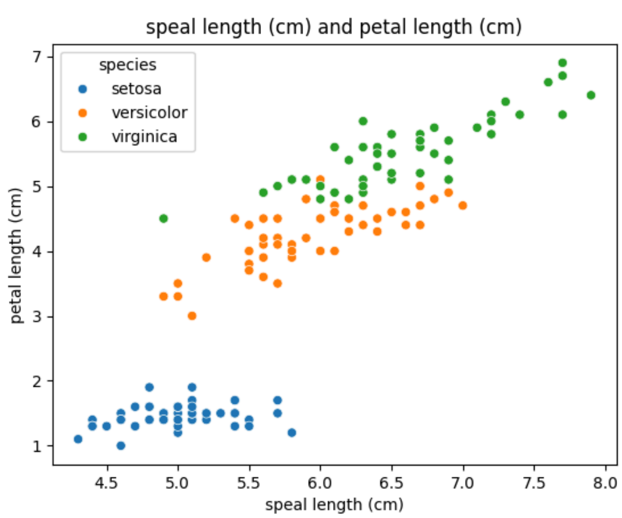


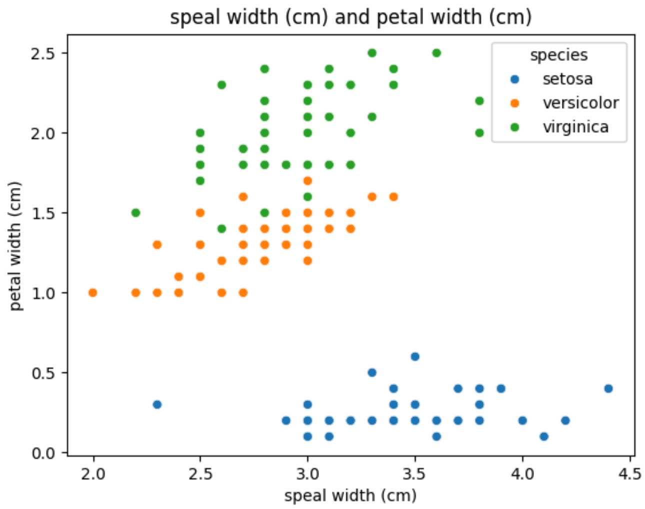
1b. Based on the histogram's density curve, I estimate that the mean and median are close to equal.
1c. mean: 3.1 median: 3.0
1d. percentile27: [2.8]

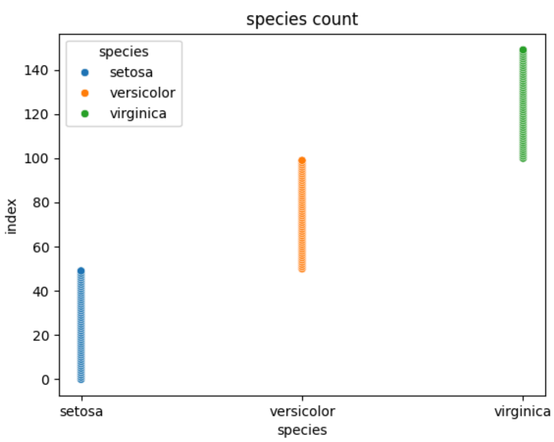
1e.


sepal length (cm) and sepal width (cm)
species
setosa
versicolor
virginica
sepal width (cm)
sepal length (cm)
4.5
4.0
3.5
3.0
2.5
2.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0


petal length (cm) and petal width (cm)
species
setosa
versicolor
virginica
petal width (cm)
petal length (cm)
2.5
2.0
1.5
1.0
0.5
0.0
1
2
3
4
5
6
7

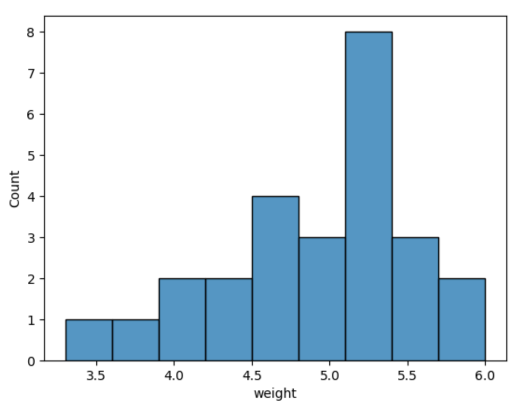

speal length (cm) and petal length (cm)
species
setosa
versicolor
virginica
petal length (cm)
speal length (cm)
7
6
5
4
3
2
1
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0



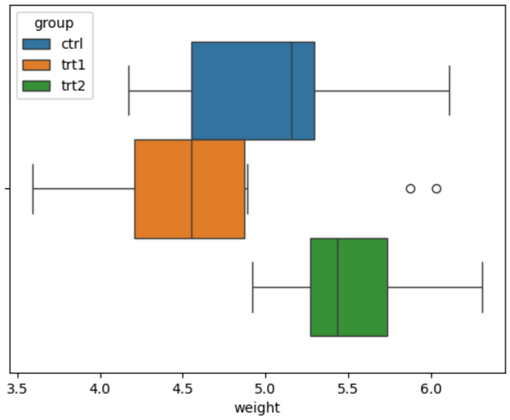


1f. The petal length and sepal length for Vigincia and Verisocolor most heavily correlate. All values for setosa have the weakest correlation for all comparisons.

2a.



2b.



2c. About 90% of trt1 values are below the minimum of trt2.
2d. 80% of the trt1 values are less than trt2's minimum
2e.